Tossaporn (Tree) Saengja

Experienced in developing advanced deep learning models (generative AI, LLMs) and leading research over five years.

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education

Massachusetts Institute of Technology
Master of Engineering in Computer Science and Engineering — 4.8/5.0 — 2019-2020
Bachelor of Science in Computer Science and Engineering — 4.8/5.0 — 2015-2019
International Olympiad in Informatics 2013 (IOI), Bronze, 85th among 299 contestants (77 countries)
ACM Boston Preliminary 2016 (BOSPRE), 4^{th} place among 37 teams (16 schools)
Teaching Assistant for MIT 6.824: Distributed Systems (Prof. Robert Morris, Spring 2020)

experience

SiData+, Siriraj Hospital — Data Scientist — 2024-Present
- Developing a novel medical reasoning Large Language Model (LLM) with SCB 10x, focusing on improving diagnostic accuracy for general medical applications.

PreceptorAI, CARIVA — AI Consultant — 2023-Present
- Developed machine learning models in collaboration with Siriraj Radiology Department:
  - Multi-label segmentation model (nnUNet, 0.98 Dice) for vertebrae L3 (muscle, fats) quantification in MRI.
  - Vision-Large Language Model to generate diagnostic reports from chest X-ray images (PyTorch).
  - Generative text-to-image model (diffusion model) for synthetic chest X-ray generation (PyTorch, MONAI).
- Led research and development of Gindee, a VLM based nutrition estimator.

Vision and Learning Lab, VISTEC — Research Assistant, under Prof. Supasorn Suwajanakorn — 2023-2024
- Investigated object collocation by developing Score Distillation Sampling (SDS) to leverage Stable Diffusion's priors.
- Enhanced local attention masking in CLIP for zero-guidance segmentation.
- Conducted experiments on diffusion model memorization (CIFAR-10).

Facebook, Notification Backend — Intern — 2017
- Implemented a pipeline for network effect prediction (classification/regression) using HiveQL and FBLearner Flow.
- Increased comment metrics by 0.5% in production (2B active users).

MIT Media Lab, Human Dynamics — Undergrad Researcher, under Yan Leng — 2018-2020
- Developed an optimization framework (Python, Pandas) for learning network structure and marginal benefits from large-scale datasets (e.g., 400GB+ CDRs).

publications

LUSD: Localized Update Score Distillation for Text-Guided Image Editing *(co-first, under review)*
W. Chinchuthakun*, **T. Saengja***, N. Tritrong, P. Rewatbowornwong, P. Khungurn, S. Suwajanakorn. arXiv:2503.11054

ZeroGuideSeg++: Improving Local Attention Masking in CLIP for Zero-Guidance Segmentation
P. Rewatbowornwong, N. Chatthee, **T. Saengja**, E. Chuangsuwanich, S. Suwajanakorn. *(under review)*

GPU Poor's Guide to Radiology Report Generation
K. Udomlapsakul, et al. (**T. Saengja** as third author). Proc. BioNLP 2024.

selected academic projects

Accurate Clinical Decision Support Systems (MIT 6.S897: ML for Healthcare, 2017)
Analyzed drivers of clinical decision support dismissals using logistic regression, random forests, LDA.
Paraphrase Identification (MIT 6.867: Machine Learning, 2016)
Designed BoW, word-embedding, SVM, NN, and LSTM models achieving 81.8% F1-score on MSR Paraphrase Corpus.

additional experience

Mahidol Wittayanusorn High School — Computer Science Teacher — 2020-2023
- Coached students to represent Thailand in IOI 2022.
- Lectured data science (Python), web dev, blockchain, Computer Olympiad (C++), programming (Python).
- Advised year-long student projects, primarily in machine learning.

Agnos — Technical Cofounder — 2019-2023
- Co-developed an AI-powered diagnosis system (Python, Django) used by physicians. (130k+ records, 50k+ active users)
- Led Data Science, AI, and Growth teams, helped securing 30M THB funding (Shark Tank Thailand S3).